

LOUISVILLE BOWMAN, KY, USA

WMO: 724235

Lat: 38.228N

Lon: 85.664W

Elev: 540

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	15.6	-2.2	5.0	13.6	2.7	6.4	17.6	21.5	43.4	19.3	39.9	6.4	310	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	93.3	75.2	91.2	74.7	89.5	73.8	78.5	88.4	77.4	87.2	76.3	85.4	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	136.5	83.3	74.6	132.1	82.5	73.2	125.9	81.4	42.4	88.6	41.1	87.0	40.0	85.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.4	14.2	4.2	97.4	7.5	3.6	-1.2	100.0	-5.5	102.1	-9.7	104.1	-15.1	106.7
			3.0	80.7	7.1	2.2	-2.2	82.3	-6.3	83.5	-10.3	84.8	-15.5	86.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	34.8	38.8	47.2	58.2	67.3	75.4	78.4	77.7	70.9	59.2	47.6	39.1	(6)
	DBStd	17.73	12.08	11.35	10.59	9.09	7.69	5.45	4.67	4.67	6.99	8.69	9.27	10.43	(7)
	HDD50	1584	493	343	182	29	1	0	0	0	0	18	153	365	(8)
	HDD65	4116	938	734	558	238	68	4	0	0	25	223	523	805	(9)
	CDD50	4499	21	29	95	275	538	763	879	860	627	303	82	27	(10)
	CDD65	1555	0	1	6	34	140	317	414	395	202	43	2	1	(11)
	CDH74	14974	0	4	51	291	1201	3014	4268	3887	1872	371	14	1	(12)
	CDH80	5694	0	0	5	43	339	1148	1788	1584	697	89	1	0	(13)
(14) Wind	WSAvg	5.8	7.3	7.2	7.0	7.0	5.8	5.1	4.6	4.3	4.3	4.9	5.9	6.4	(14)
(15) Precipitation	PrecAvg	46.0	3.1	3.3	4.6	4.4	4.8	3.9	4.3	3.5	3.5	3.1	3.6	3.9	(15)
	PrecMax	69.0	7.2	10.6	14.9	14.0	11.6	10.1	10.0	8.8	10.9	10.0	7.6	8.9	(16)
	PrecMin	32.6	0.4	0.7	1.0	0.7	1.4	0.5	0.4	0.6	0.1	0.4	0.7	0.6	(17)
	PrecStd	8.1	1.6	2.0	2.6	2.6	2.3	2.1	2.1	1.9	2.4	2.1	1.7	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.8	72.3	80.0	84.1	89.5	93.4	97.1	96.3	94.7	87.1	75.5	69.1	(19)
		MCWB	60.5	61.0	63.8	65.3	72.6	75.3	77.7	75.2	72.0	70.9	64.2	61.3	(20)
	2%	DB	63.2	66.2	74.4	81.0	86.3	91.1	93.2	93.1	90.6	82.0	71.7	63.8	(21)
		MCWB	58.0	56.8	60.4	64.1	70.3	74.7	76.5	74.9	71.8	67.5	60.4	57.8	(22)
	5%	DB	58.7	62.0	69.7	77.5	83.9	89.1	91.0	90.4	87.7	78.4	66.8	59.4	(23)
		MCWB	54.0	53.9	57.6	61.9	69.2	73.6	75.6	74.3	71.0	65.2	58.1	53.9	(24)
	10%	DB	53.9	56.8	64.7	73.8	81.3	86.7	88.7	88.1	83.9	73.9	63.1	54.9	(25)
		MCWB	48.9	49.6	55.0	60.8	68.5	72.5	74.6	73.5	69.7	63.6	55.9	49.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	63.2	65.6	69.3	75.2	78.8	81.4	80.1	76.8	74.1	66.3	63.5	(27)
		MCDB	66.1	70.6	77.1	80.0	84.3	88.6	90.9	90.1	87.7	82.6	72.0	68.1	(28)
	2%	WB	58.9	59.4	62.5	67.1	73.3	77.3	79.0	78.1	75.0	70.8	63.2	59.2	(29)
		MCDB	62.5	64.6	71.0	75.9	82.8	87.3	89.1	87.8	84.5	78.2	68.7	63.3	(30)
	5%	WB	54.6	55.1	59.9	64.9	71.6	76.1	77.8	76.9	73.4	68.0	59.8	55.0	(31)
		MCDB	58.1	60.1	67.6	72.8	81.1	85.4	87.6	86.1	82.1	75.7	65.7	58.5	(32)
	10%	WB	49.1	50.1	56.5	62.8	69.9	74.7	76.6	75.5	72.0	65.1	56.5	50.5	(33)
		MCDB	53.7	56.0	63.4	70.9	78.0	83.6	85.8	84.0	80.4	72.0	62.2	54.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	16.5	18.5	20.4	18.7	17.9	17.6	18.4	20.1	20.2	17.8	14.5	(35)
		MCDBR	19.6	22.5	24.2	24.3	21.1	20.3	19.9	21.0	24.0	24.3	22.7	19.7	(36)
		MCWBR	15.7	16.4	14.1	12.3	9.2	7.9	6.9	7.4	8.9	11.5	14.6	15.6	(37)
	5% WB	MCDBR	17.9	19.8	20.2	18.8	18.0	17.7	17.4	17.4	18.4	19.5	19.3	17.4	(38)
		MCWBR	15.5	16.9	13.7	11.7	9.2	7.9	7.6	7.3	8.3	10.9	14.8	15.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.329	0.359	0.403	0.442	0.455	0.479	0.476	0.421	0.362	0.330	0.308	(40)
	taud		2.458	2.390	2.381	2.259	2.190	2.188	2.131	2.131	2.260	2.427	2.465	2.482	(41)
	Ebn at Noon		274	284	286	278	268	263	256	253	261	269	266	269	(42)
	Edh at Noon		26	32	35	42	46	46	49	48	39	30	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		607	830	1174	1557	1737	1973	1872	1750	1440	1070	731	522	(44)
	RadStd		64	116	110	109	155	111	115	100	135	115	81	51	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+153
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+139

Nomenclature: See separate page